Topological Ramsey spaces, associated ultrafilters, and their applications to the Tukey theory of ultrafilters and Dedekind cuts of nonstandard arithmetic.

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Sesquicentennial Conference on Infinitary Ramsey Theory

University of Denver

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Motivations and Submissions

- [2] Blass, *Ultrafilter mappings and their Dedekind cuts*

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- [5] Dobrinen, Mijares and T., *Topological Ramsey Spaces from Fraïsse Classes, Ramsey-classification theorems, and initial structures in the Tukey types of p-points*

Chapter 1

§1.1 Higher dimensional Ramsey theory

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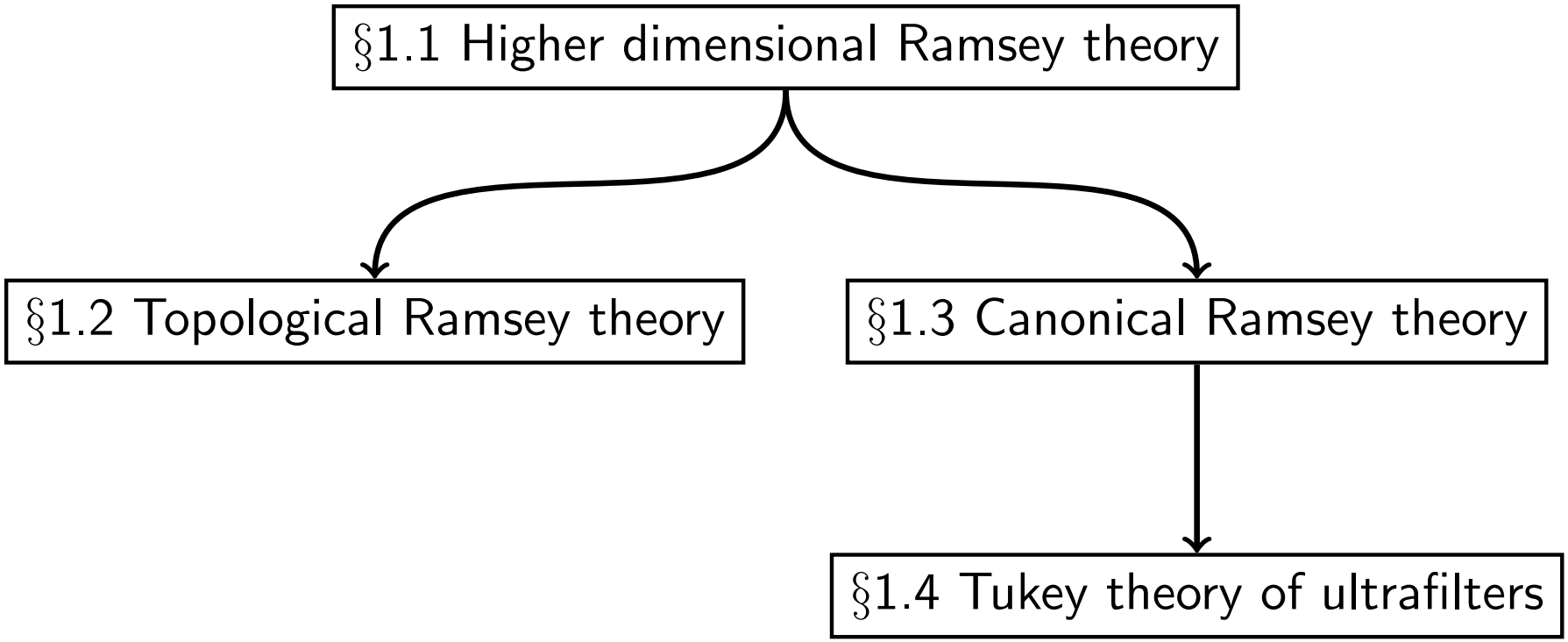
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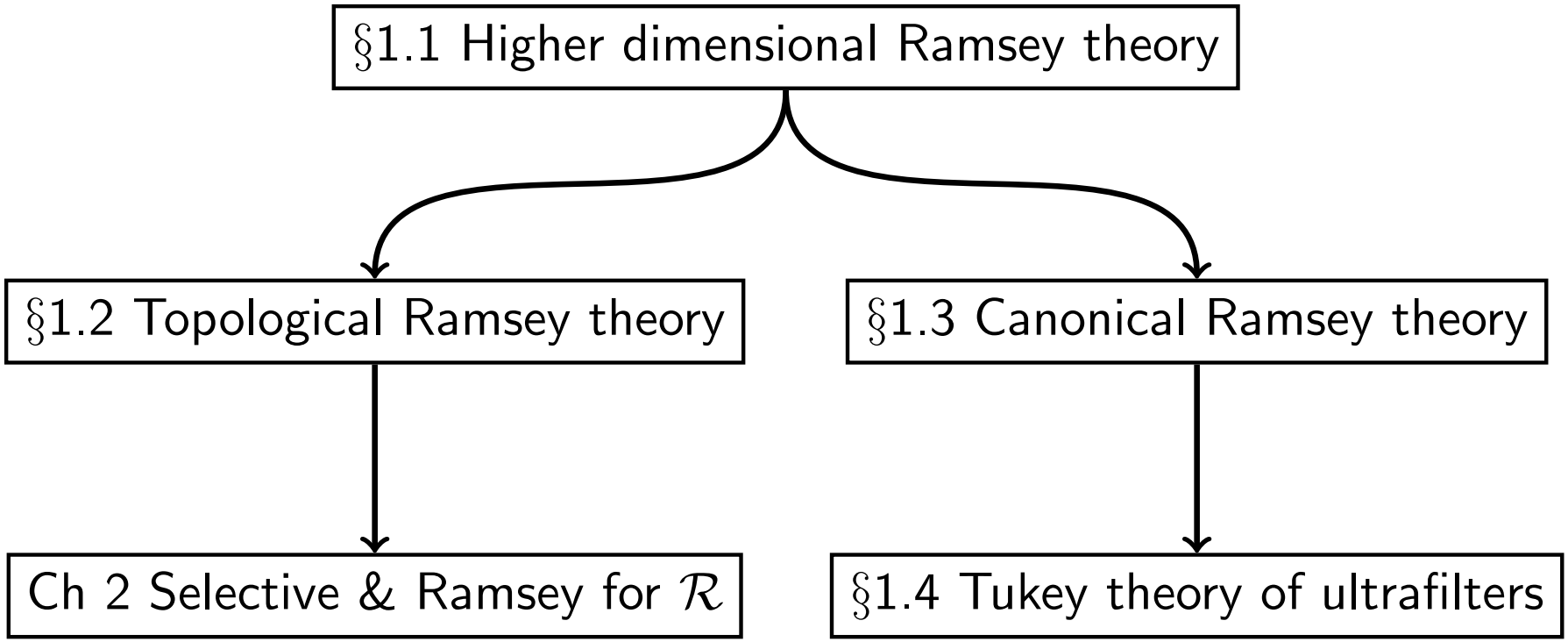
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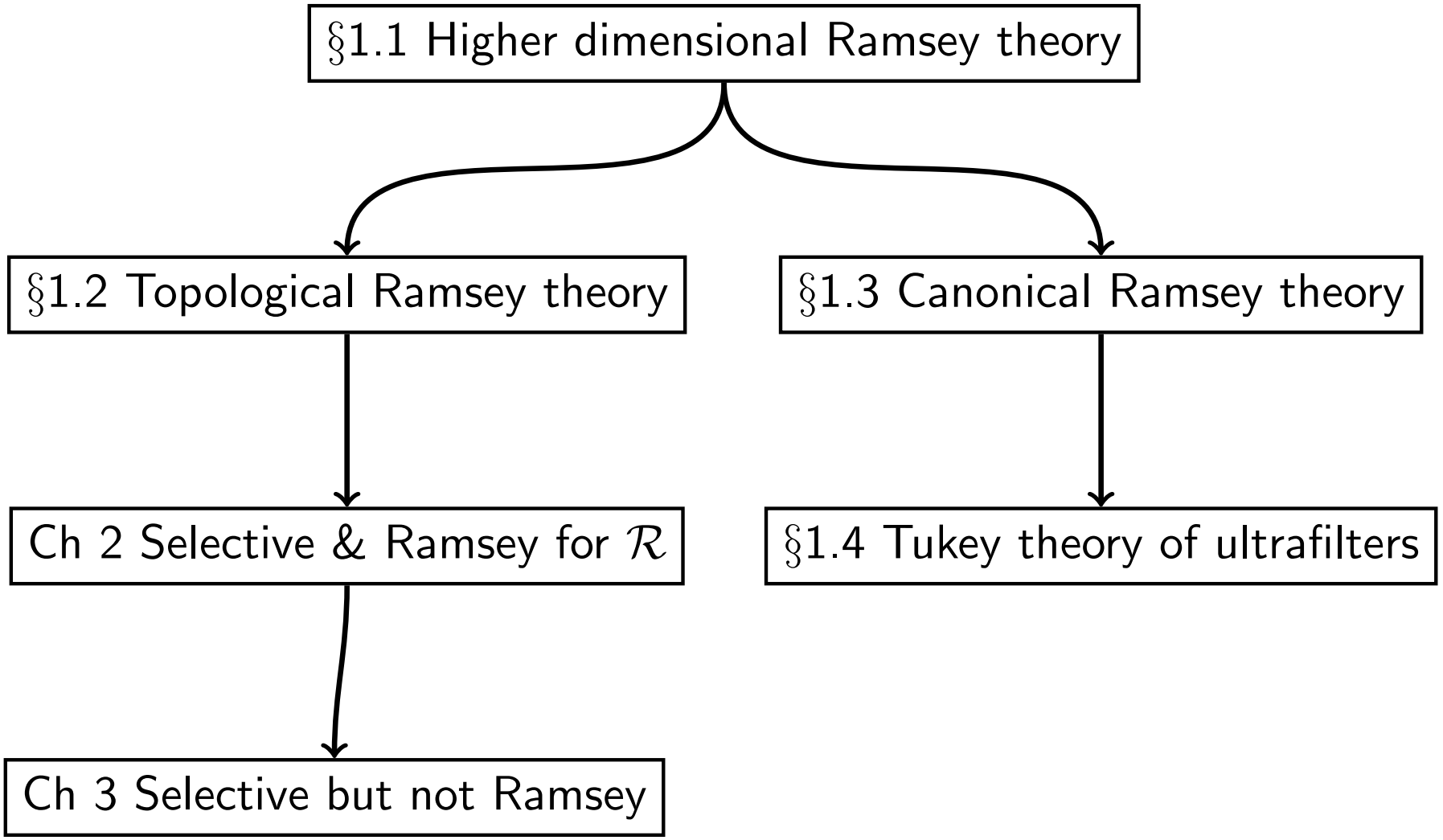
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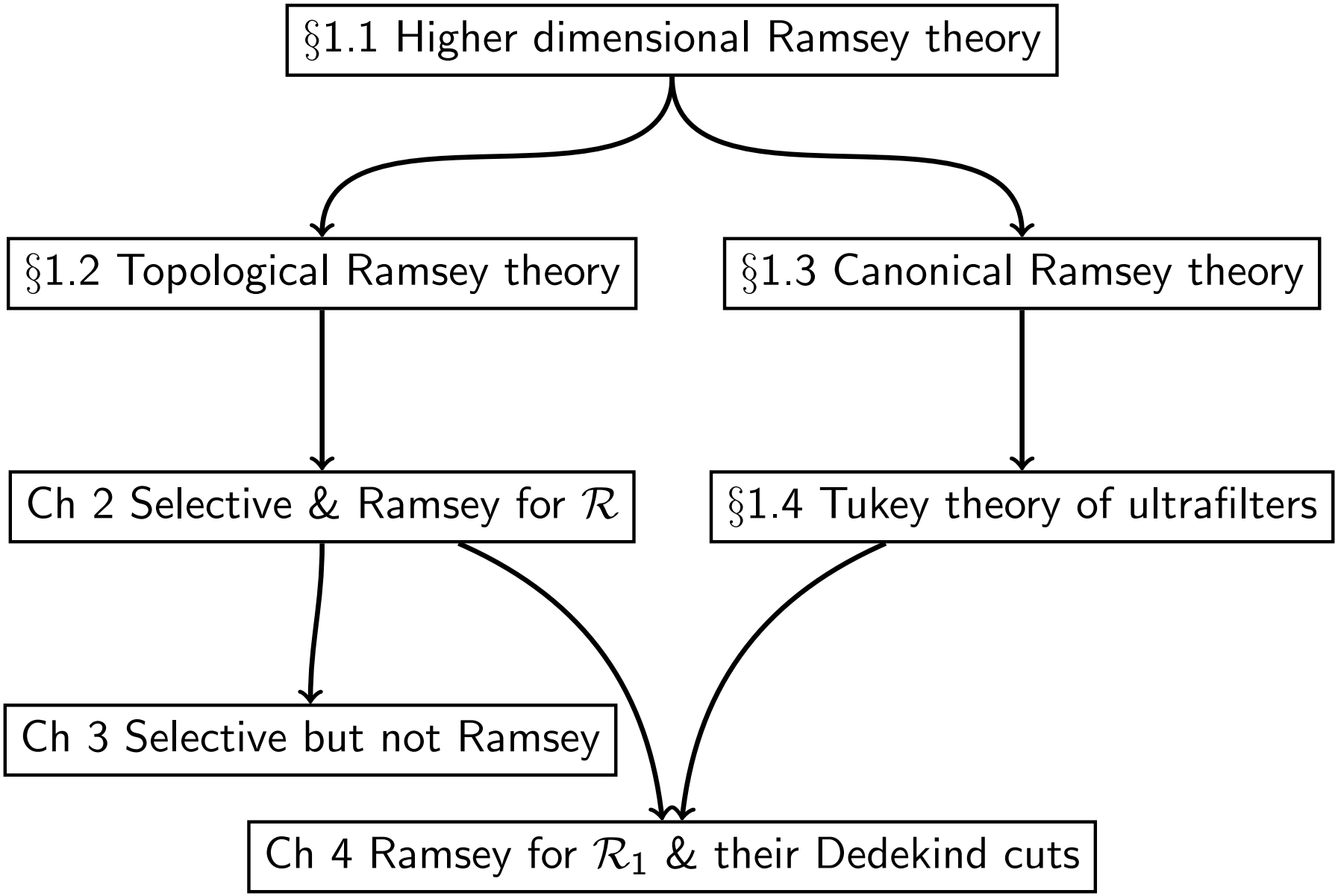
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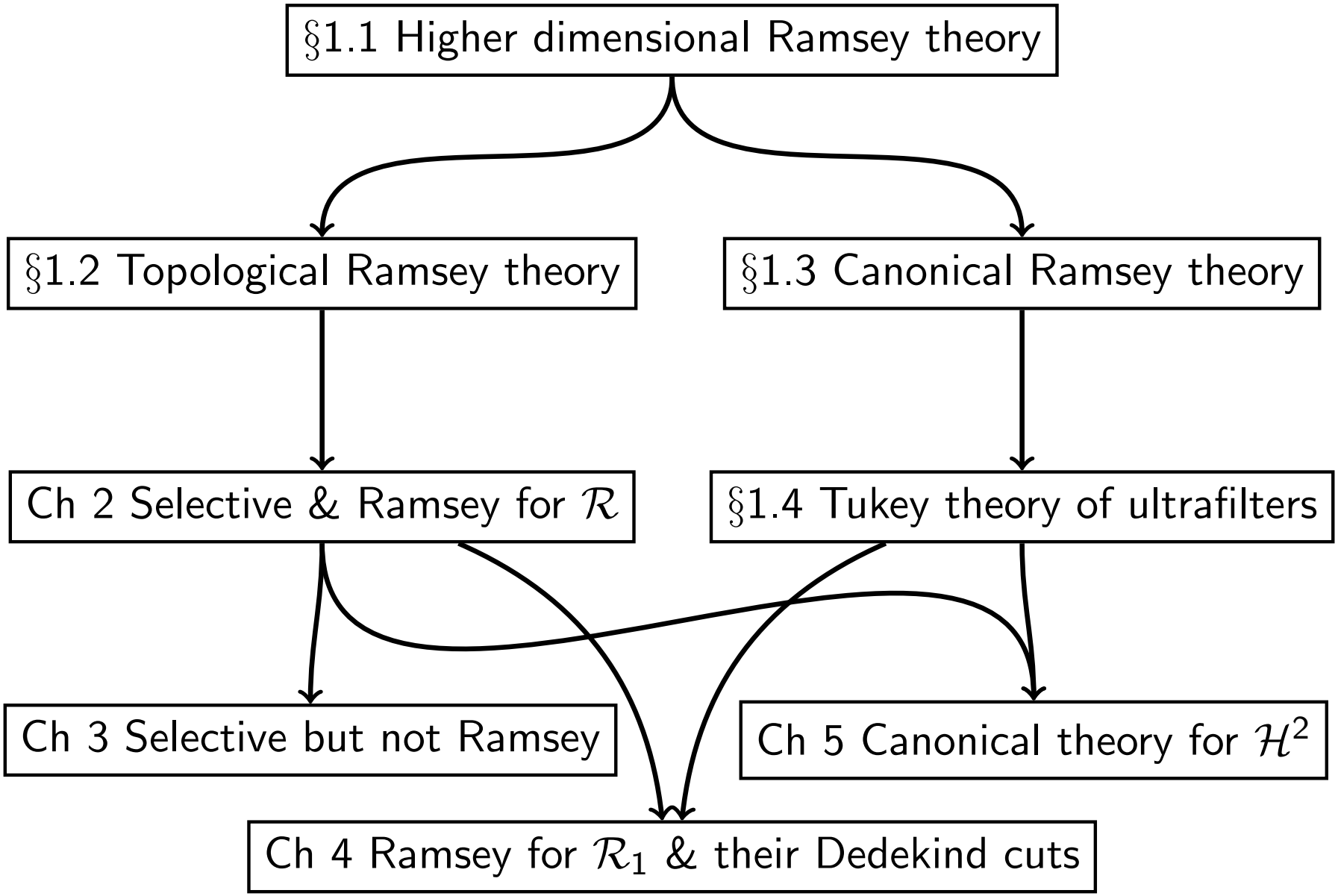
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Theorem (Kunen, [3])

$\mathcal{U}$ is Ramsey iff $\mathcal{U}$ is selective iff $\mathcal{U}$ is minimal in the Rudin-Keisler ordering.

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$$s \sqsubseteq t \text{ if and only if for each } i < |s|, s_i = t_i.$$

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- $\pi_0 : \omega^{<\omega} \to \omega^1$

$$\pi_0(s) = \begin{cases} \langle s_0 \rangle & \text{if} s \neq \langle\rangle \\ \langle\rangle & \text{otherwise.} \end{cases}$$

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Let $\mathcal{R}(T)$ denote the set of all subtrees of T isomorphic to T.

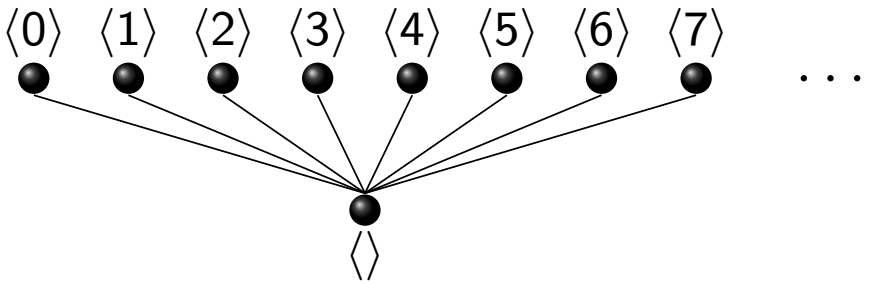

Figure: Graph of T_0

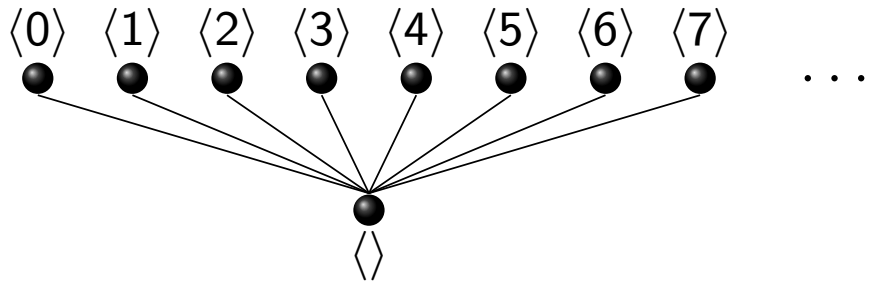

Figure: Graph of T_0

Remark

If $S \in \mathcal{R}(T_0)$ then there exists a strictly increasing sequence $(k_i)_{i<\omega}$ such that $\pi_0''[S] = \{\langle k_i \rangle : i < \omega\}$.

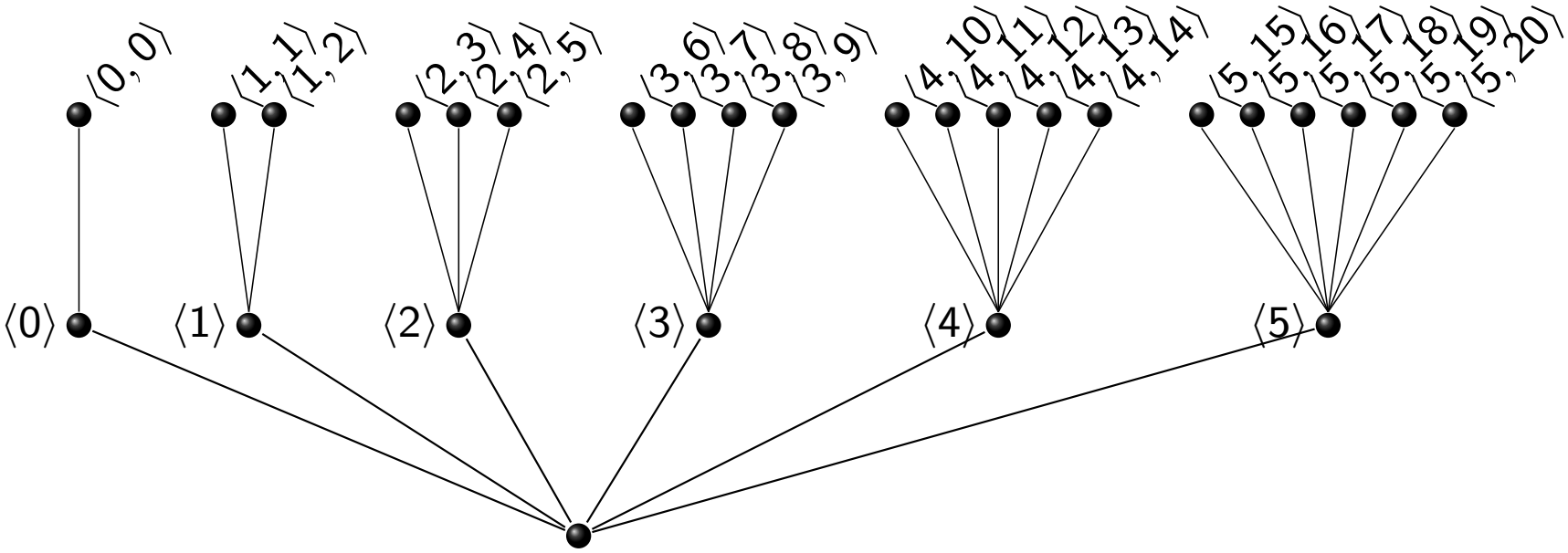

Figure: Graph of T_1

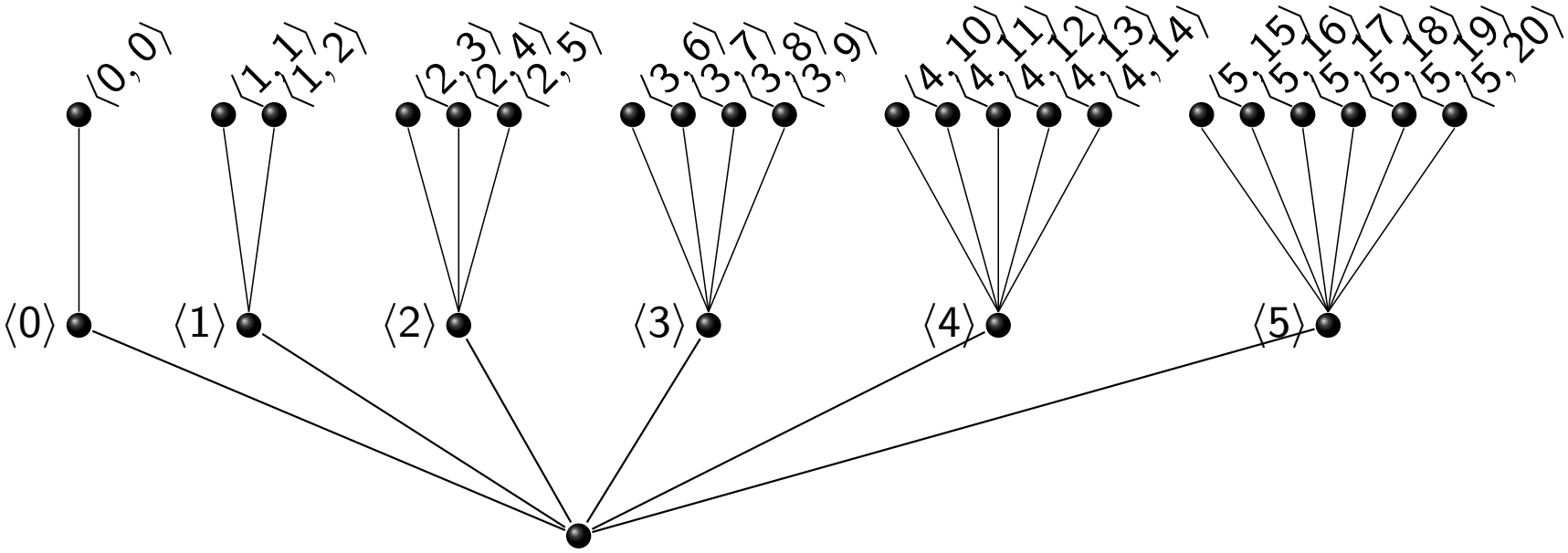

Figure: Graph of T_1

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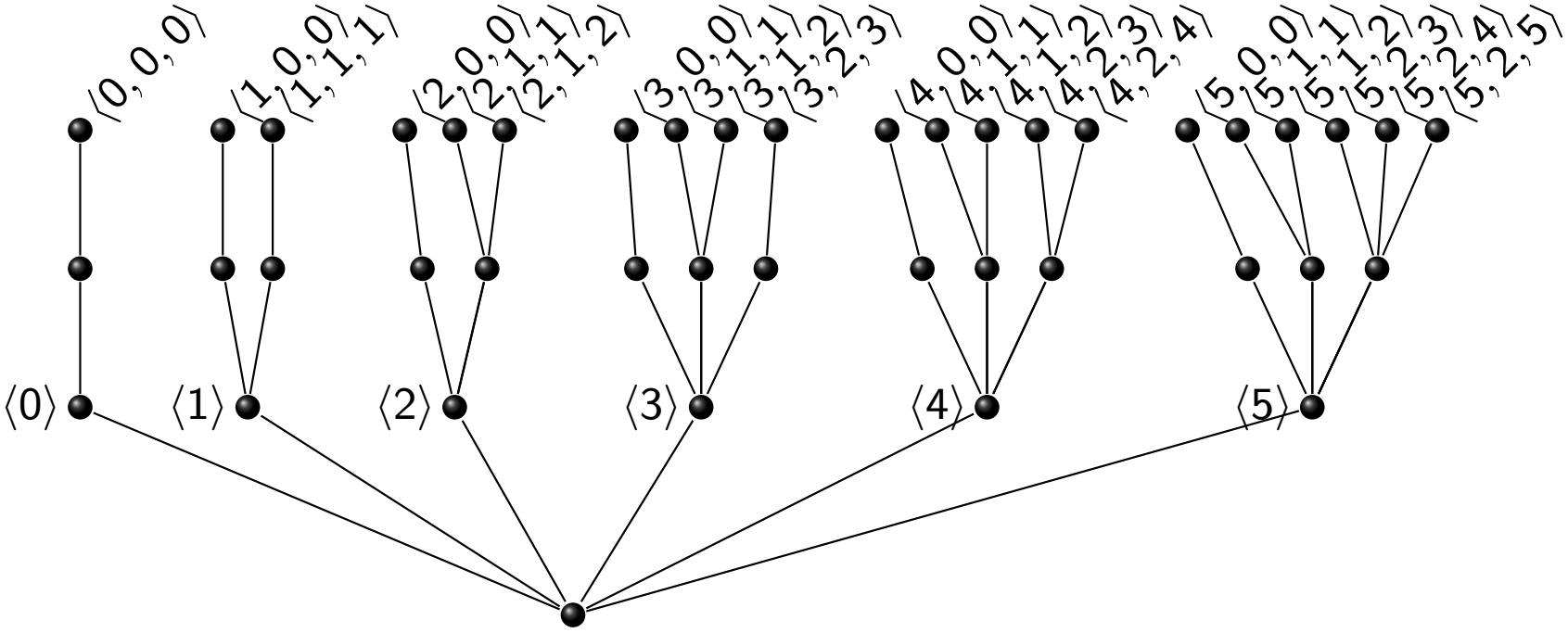
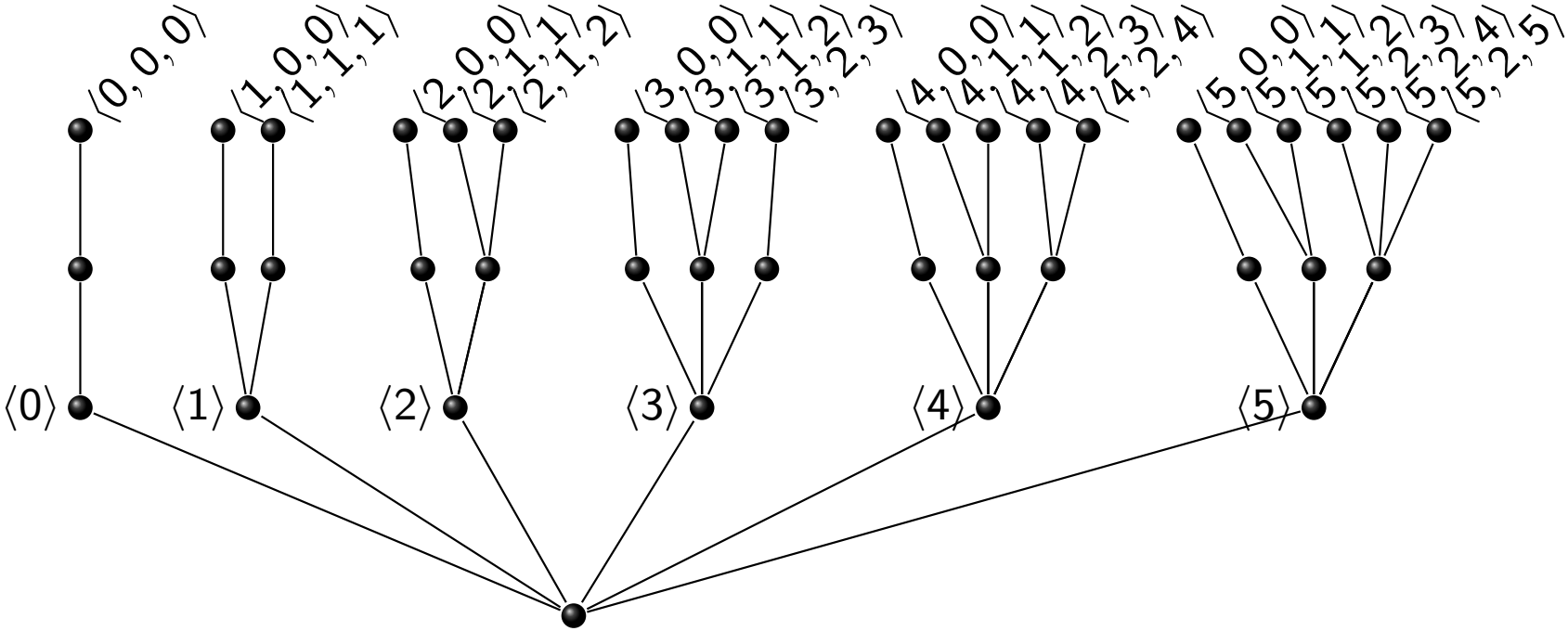


Figure: Graph of the tree $T_1^\star$



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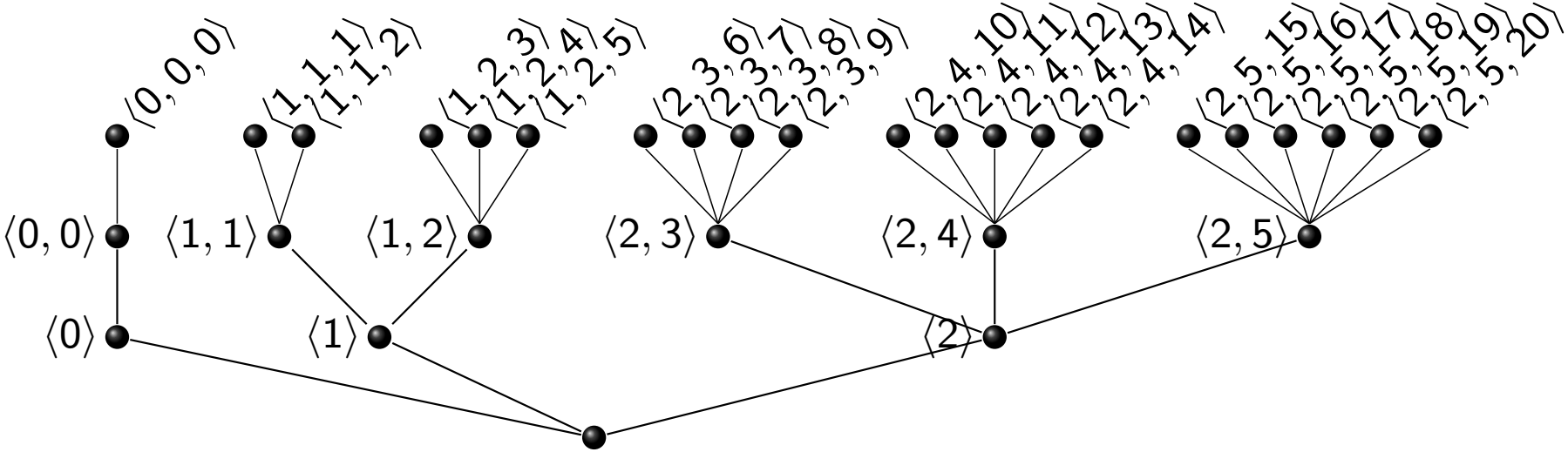
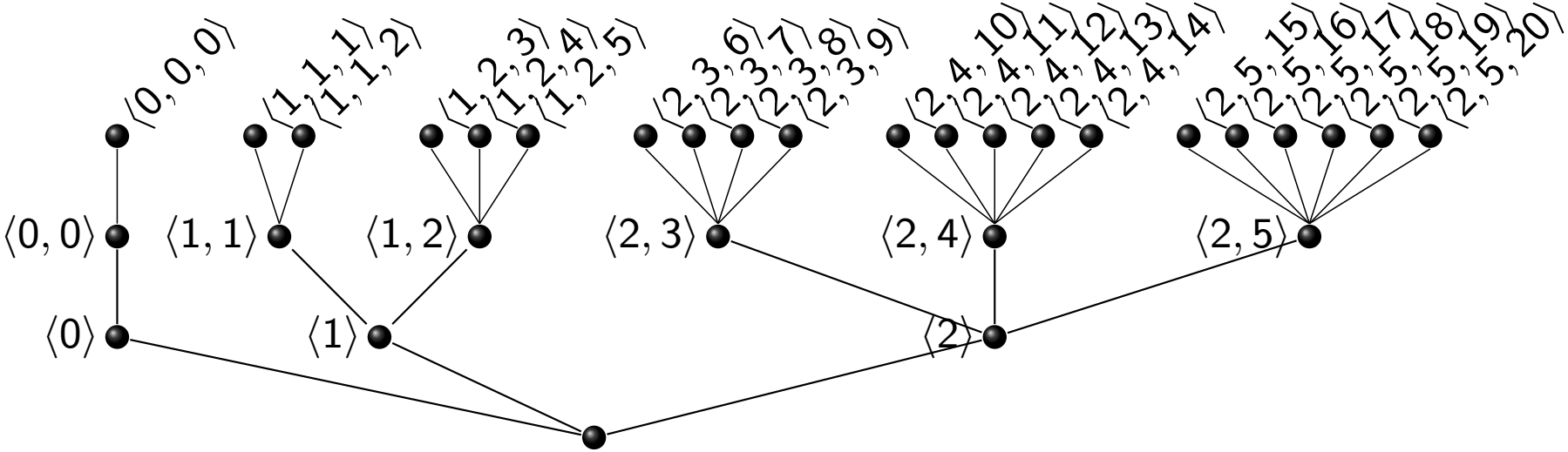


Figure: Graph of T_2



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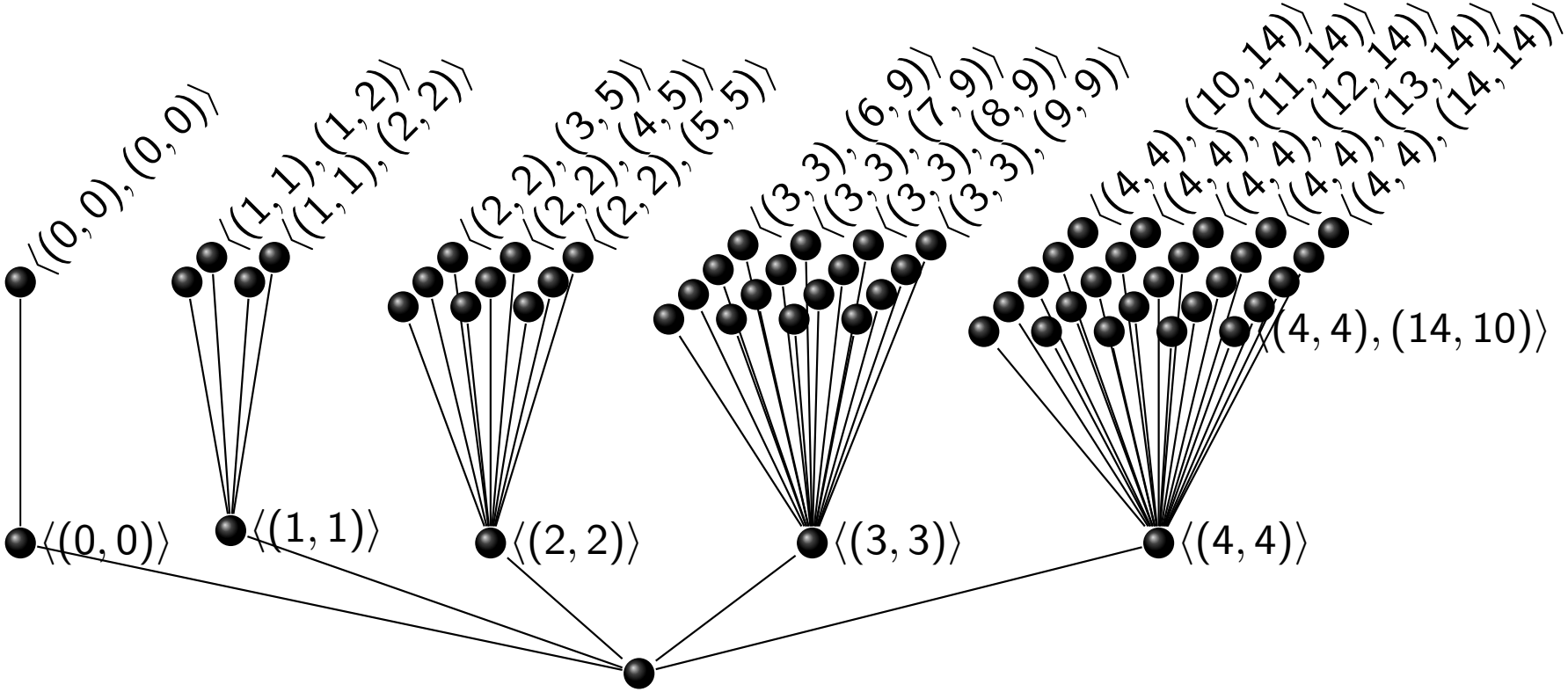
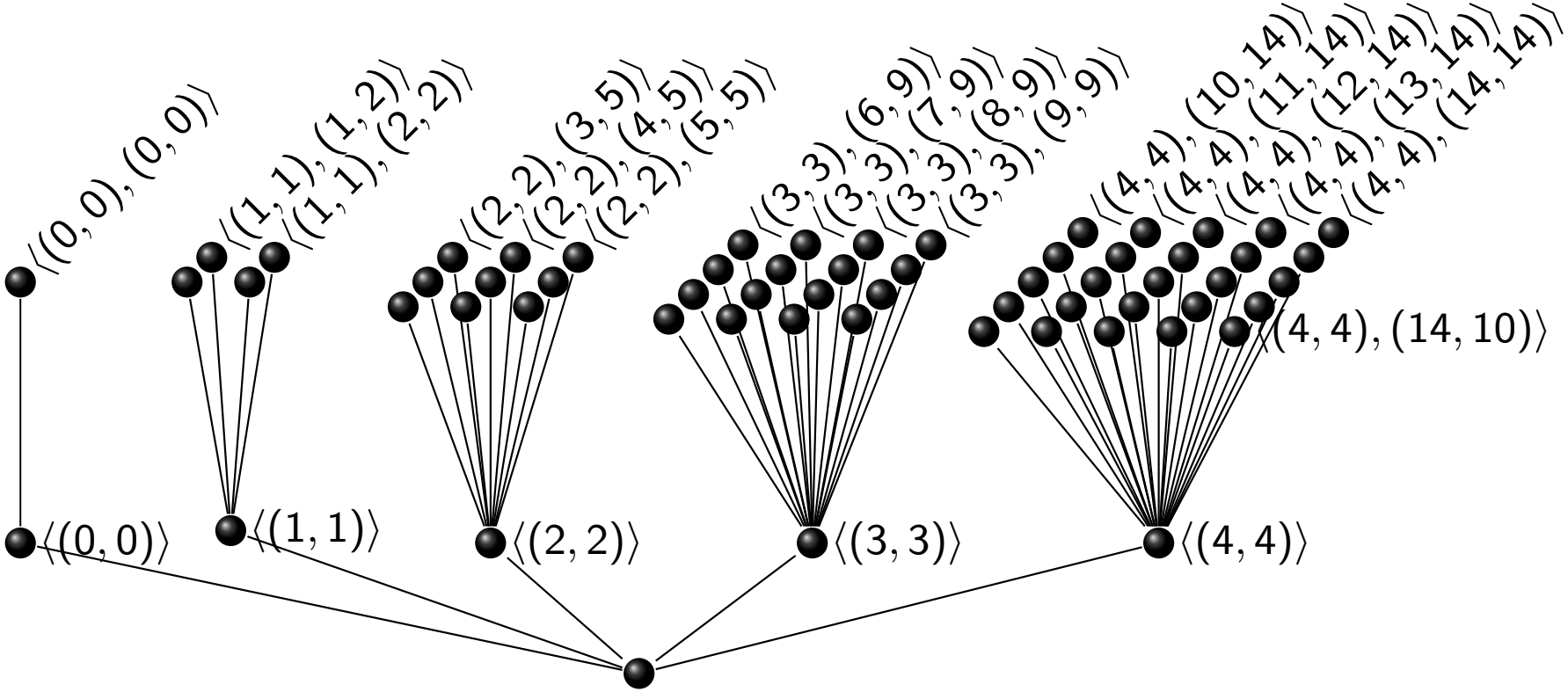


Figure: Graph of the tree $T_1 \otimes T_1$



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Remark

If $S \in \mathcal{R}(T_1 \otimes T_1)$ then there exists a strictly increasing sequence $(k_i)_{i<\omega}$ such that $\pi_0''[S] = \{\langle (k_i, k_i) \rangle : i < \omega\}$.

Definition (T., [13])

For each $S \in \mathcal{R}(T)$ there is a strictly increasing sequence $(k_i)_{i<\omega}$ such that $\pi_0''[S] = \{\langle (k_i, \ldots, k_i) \rangle : i < \omega\}$.

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For $S, S' \in \mathcal{R}(T)$, $S \leq S'$ if and only if S is subtree of S'.

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Definition (Mijares, [10])

The **almost-reduction relation** is defined as follows: for $S, S' \in \mathcal{R}(T)$, $S \leq^* S'$ if and only if there exists $i < \omega$ such that $S \setminus r_i(S) \subseteq S'$.

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Notation

$\mathcal{R}_n := \mathcal{R}(T_n)$, $\mathcal{R}_n^\star := \mathcal{R}(T_n^\star)$, $\mathcal{H}^2 := \mathcal{R}(T_1 \otimes T_1)$

Definition (Mijares, [10])

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- An ultrafilter $\mathcal{U}$ generated by $\mathcal{G} \subseteq \mathcal{R}(T)$ is **selective for** $\mathcal{R}(T)$ if and only if for each decreasing sequence $S_0 \geq S_1 \geq S_2 \geq \ldots$ of elements of $\mathcal{G}$, there exists another $S \in \mathcal{G}$ such that for all $i < \omega$, $S \setminus r_i(S) \subseteq S_i$.

Definition (Mijares, [10])

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Theorem (Mijares, [10])

Ramsey for $\mathcal{R}(T) \Rightarrow$ selective for $\mathcal{R}(T)$.

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$(\mathcal{R}_1^\star, \leq^*)$ is σ-closed and there exists a map $\Gamma : \mathcal{R}_1^\star \to \mathcal{R}_1$ such that if $\mathcal{G}$ is a generic filter for $(\mathcal{R}_1^\star, \leq^*)$ over some ground model V, then $\Gamma''\mathcal{G}$ generates an ultrafilter on $[T_1]$ that is selective for $\mathcal{R}_1$ but not Ramsey for $\mathcal{R}_1$ in $V[\mathcal{G}]$.

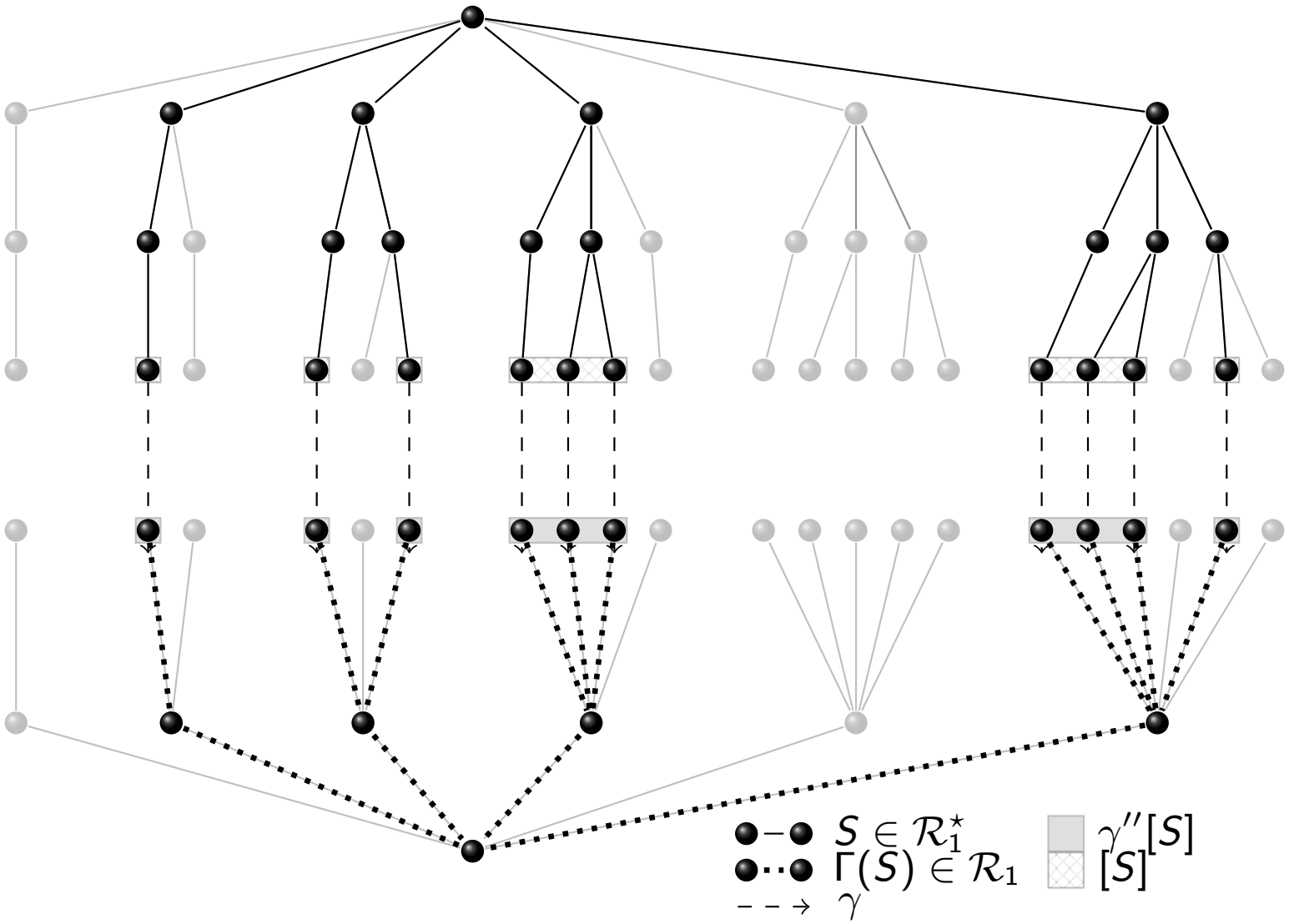

Figure: Graph of $S \in \mathcal{R}_1^\star$ and $\Gamma(S) \in \mathcal{R}_1$.

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Theorem (T., [13])

Suppose that $\langle S_i : i \leq n \rangle$ is a finite sequence of trees where each S_i is one of the trees T_j for some $j < \omega$. $(\bigotimes_{i=0}^n \mathcal{R}(S_i^\star), \leq^*)$ is σ-closed and there exists a map $\mathbf{\Gamma} : \bigotimes_{i=0}^n \mathcal{R}(S_i^\star) \to \bigotimes_{i=0}^n \mathcal{R}(S_i)$ such that if $\mathcal{G}$ is a generic filter for $(\bigotimes_{i=0}^n \mathcal{R}(S_i^\star), \leq^*)$ over some ground model V, then $\mathbf{\Gamma}''\mathcal{G}$ generates an ultrafilter on $[\bigotimes_{i=0}^n S_i]$ that is selective for $\bigotimes_{i=0}^n \mathcal{R}(S_i)$ but not Ramsey for $\bigotimes_{i=0}^n \mathcal{R}(S_i)$ in $V[\mathcal{G}]$.

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$$f(\mathcal{U}) = \left\langle \{f''Z : Z \in \mathcal{U}\} \right\rangle . \tag{2}$$

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A **Rudin-Keisler mapping from** $\mathcal{U}$ **to** $\mathcal{V}$ is a function $f : X \to Y$ such that $\mathcal{V} = f(\mathcal{U})$.

Definition (Blass, [2])

- Associated to each Rudin-Keisler mapping from $\mathcal{U}$ on X to $\mathcal{V}$ on Y is a Dedekind cut in the ultrapower $\omega^Y/\mathcal{V}$.

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- The set of all equivalence classes of cardinality functions of sets in $\mathcal{U}$, and all larger elements of $\omega^Y/p(\mathcal{U})$, constitute the upper part L of a cut (S, L) of $\omega^Y/p(\mathcal{U})$, which we call **the cut associated to p and $\mathcal{U}$**.

Lemma (T., [12])

Assume the Continuum Hypothesis. If $\mathcal{V}$ is selective then there exists a Ramsey for $\mathcal{R}_1$ ultrafilter $\mathcal{U}$ such that the cut associated to $\mathcal{U}$ and π is the standard cut in $\omega^\omega/\mathcal{V}$.

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Lemma (T., [12])

Let $\mathcal{U}$ be a Ramsey for $\mathcal{R}_1$ ultrafilter on $[T_1]$ and p be a map from $[T_1]$ to ω. The cut associated to p and $\mathcal{U}$ is the standard cut in $\omega^\omega/p(\mathcal{U})$.

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Theorem (T., [12])

Assume the Continuum Hypothesis. (S, L) is the cut associated to some map of some Ramsey for $\mathcal{R}_1$ ultrafilter on $[T_1]$ to $\mathcal{V}$ if and only if $\mathcal{V}$ is selective and (S, L) is the standard cut in $\omega^\omega/\mathcal{V}$.

Definition

Let $\mathcal{U}$ be a nonprincipal ultrafilter on ω.

$\mathcal{U}$ is a **p-point** if for each $f : \omega \to \omega$ there exits $X \in \mathcal{U}$ such that $f \restriction X$ is constant or $f \restriction X$ is finite-to-one.

$\mathcal{U}$ is **weakly-Ramsey** if for each $F : [\omega]^2 \to 3$ there exits $X \in \mathcal{U}$ such that F omits a value on $[X]^2$.

Corollary (T., [12])

Assume the Continuum Hypothesis. There is a weakly-Ramsey ultrafilter on $[T_1]$ that is not Ramsey for $\mathcal{R}_1$.

Definition (Tukey, [14])

Suppose that $\mathcal{U}$ and $\mathcal{V}$ are ultrafilters on the base sets X and Y respectively.

- A function f from $\mathcal{U}$ to $\mathcal{V}$ is **cofinal** if every cofinal subset of $(\mathcal{U}, \supseteq)$ is mapped by f to a cofinal subset of $(\mathcal{V}, \supseteq)$.(In other words, f maps filter bases of $\mathcal{U}$ to filter bases of $\mathcal{V}$.)

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- The relation $\equiv_T$ is an equivalence relation and $\leq_T$ is a partial order on its equivalence classes.
- The equivalence classes are also called **Tukey types**.

Theorem (T., [12])

Suppose $\mathcal{U}$ is a Ramsey for $\mathcal{R}_1$ ultrafilter on $[T_1]$. If (S, L) is the cut associated to some map from some p-point ultrafilter in the Tukey type of $\mathcal{U}$ to some ultrafilter $\mathcal{V}$, then $\mathcal{V}$ is a p-point ultrafilter and (S, L) is the standard cut.

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Corollary (T., [12])

Assume the Continuum Hypothesis holds and $\mathcal{U}$ is a Ramsey for $\mathcal{R}_1$ ultrafilter on $[T_1]$. There exists a weakly Ramsey ultrafilter $\mathcal{V}$ such that $\mathcal{V} \not\leq_T \mathcal{U}$.

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Corollary (T., [12])

Assume the Continuum Hypothesis holds and $\mathcal{U}$ is a Ramsey for $\mathcal{R}_1$ ultrafilter on $[T_1]$. There exists a p-point ultrafilter $\mathcal{W}$ which is not weakly Ramsey such that $\mathcal{W} >_T \mathcal{U}$.

Definition (T., [11])

- Let $\tilde{S}(n)$ denote the tree $\{\langle\ \rangle, \langle 0\rangle, \langle 0, (i,j)\rangle : (i,j) \in (n+1) \times (n+1)\}$.

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$$\begin{cases} \{\langle\ \rangle, \langle 0\rangle, \langle 0,(i,\cdot)\rangle : i \in I\} & \text{if } I \neq \emptyset \ \&\ J = \emptyset, \\ \{\langle\ \rangle, \langle 0\rangle, \langle 0,(\cdot,j)\rangle : j \in J\} & \text{if } I = \emptyset \ \&\ J \neq \emptyset, \\ \{\langle\ \rangle, \langle 0\rangle, \langle 0,(i,j)\rangle : (i,j) \in I\times J\} & \text{if } I \neq \emptyset \ \&\ J \neq \emptyset. \end{cases}$$

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- Let $\mathcal{T}(n)$ denote the collection of trees of the form $T_{\langle\rangle}$, $T_{\langle 0\rangle}$ and $T(I,J)$ with $I, J \subseteq n+1$ and either I or J nonempty.

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- With $L = \{l_0, \ldots, l_n\}$ and $K = \{k_0, \ldots, k_n\}$.
- The T-**projection of** $X(n)$ denoted by $\pi_T(X(n))$ is given by

$$\begin{cases} \{\langle\ \rangle\} & \text{if } T = T_{\langle\rangle}, \\ \{\langle\ \rangle, \langle m \rangle\} & \text{if } T = T_{\langle 0 \rangle}, \\ \{\langle\ \rangle, \langle m \rangle, \langle m, k_j \rangle : j \in J\} & \text{if } T = T(\emptyset, J), \\ \{\langle\ \rangle, \langle m \rangle, \langle m, l_i \rangle : i \in I\} & \text{if } T = T(I, \emptyset), \\ \{\langle\ \rangle, \langle m \rangle, \langle m, (l_i, k_j) \rangle : (i, j) \in I \times J\} & \text{otherwise.} \end{cases}$$

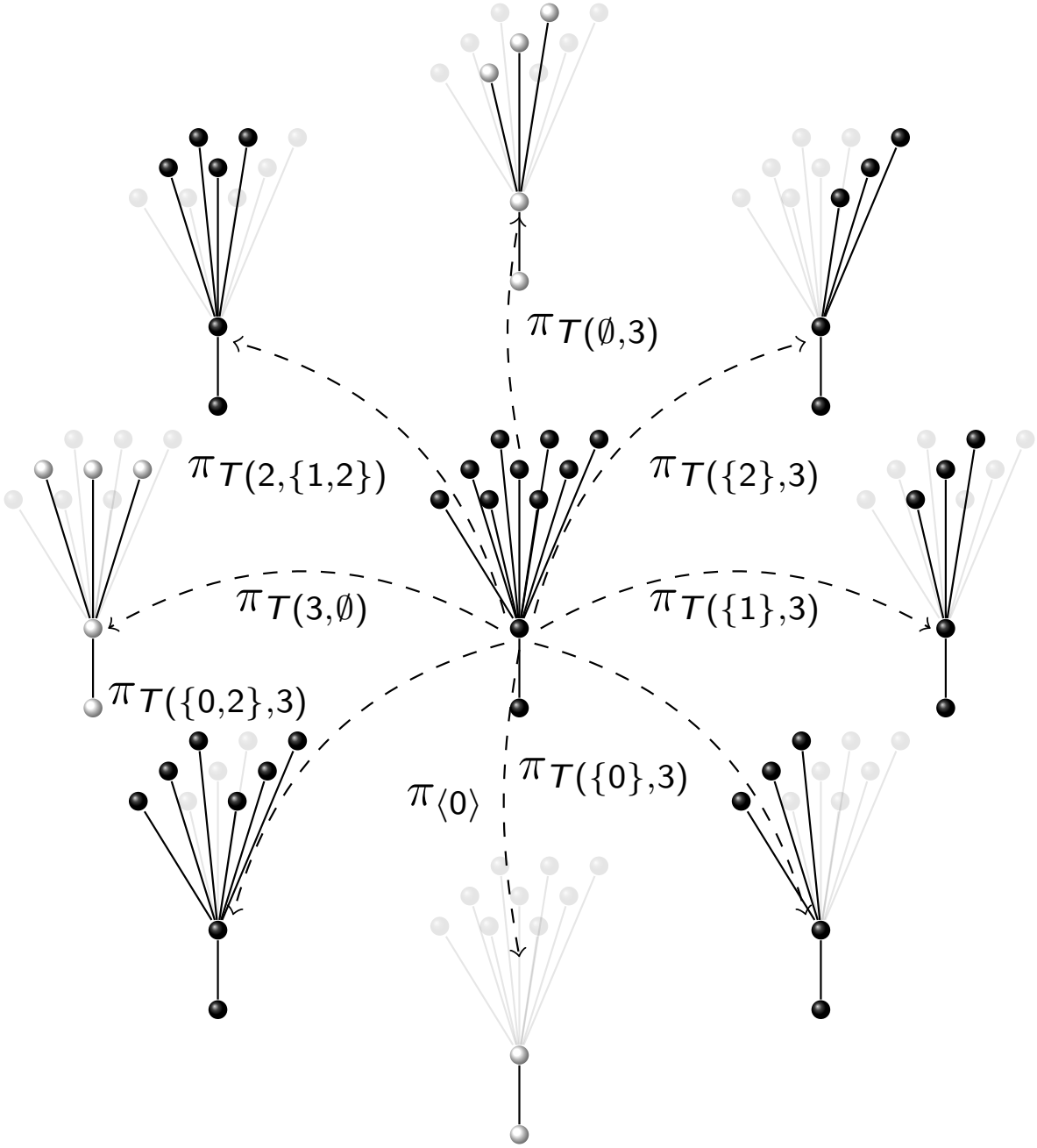

Figure: Graphs of various projections of an element of $\mathcal{H}^2(2)$

Definition (T., [11])

- Each $T \in \mathcal{T}(n)$ induces an equivalence relation E_T, on $\mathcal{H}^2(n)$ by

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Theorem (Erdős-Rado Theorem for $\mathcal{H}^2$; T., [11])

For each $A \in \mathcal{H}^2$ and equivalence relation E on $\mathcal{H}^2(n)|A$, there exists $B \leq A$ and $E_T \in \mathcal{E}(n)$ such that

$$E_T \restriction (\mathcal{H}^2(n)|B) = E \restriction (\mathcal{H}^2(n)|B).$$

Definition

A subset $\mathcal{F}$ of $\mathcal{AH}^2$ is a **front on** $\mathcal{H}^2$ if for each $X \in \mathcal{H}^2$ there exists $a \in \mathcal{F}$ and $i < \omega$ such that $a = r_i(X)$ and $b \not\sqsubseteq c$ for all $b \neq c \in \mathcal{F}$.

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Definition

Let $\mathcal{F}$ be a front on $\mathcal{H}^2$ and let φ be a function on $\mathcal{F}$.

- φ is **inner** if for each $a \in \mathcal{F}$ there exists a family of trees $\{T_i : i < |a|\}$ such that for each $i < \omega$ $T(i) \in \mathcal{T}(i)$ and

$$\varphi(a) = \bigcup_{i<|a|} \pi_{T_i}(a(i)).$$

- φ is **Nash-Williams** if $\varphi(a) \not\sqsubseteq \varphi(b)$, for all $a \neq b \in \mathcal{F}$.

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Theorem (T., [11])

Suppose $A \in \mathcal{H}^2$, $\mathcal{F}$ is a front and R is an equivalence relation on $\mathcal{F}$. Then there is a $B \leq A$ such that R is canonical on $\mathcal{F}|B$.

Proposition (T., [11])

Assume that $\mathcal{U}$ is a Ramsey for $\mathcal{H}^2$ ultrafilter on $[T_1 \otimes T_1]$ and generated by $\mathcal{C} \subseteq \mathcal{H}^2$. Suppose $\mathcal{C}$ has basic Tukey reductions and $\mathcal{V}$ is a nonprincipal ultrafilter on ω with $\mathcal{U} \geq_T \mathcal{V}$. Then there is a front $\mathcal{F}$ on $\mathcal{C}$ and a function $f : \mathcal{F} \to \omega$ such that $\mathcal{V} = f(\langle \mathcal{C} \restriction \mathcal{F} \rangle)$.

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If $\mathcal{C} \subseteq \mathcal{H}^2$ generates a selective for $\mathcal{H}^2$ ultrafilter on $[T_1 \otimes T_1]$ then $\mathcal{C}$ has basic Tukey reductions.

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Theorem (T., [11])

If $\mathcal{C} \subseteq \mathcal{H}^2$ generates a Ramsey for $\mathcal{H}^2$ ultrafilter on $[T_1 \otimes T_1]$ then for any front $\mathcal{F}$ on $\mathcal{R}$ and any equivalence relation R on $\mathcal{F}$, there exists a $C \in \mathcal{C}$ such that R is canonical on $\mathcal{F}|C$.

Definition (T., [11])

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Theorem (T., [11])

Suppose that $\mathcal{U}$ is a Ramsey for $\mathcal{H}^2$ ultrafilter on $[T_1 \otimes T_1]$ generated $\mathcal{C} \subseteq \mathcal{H}^2$. If $\mathcal{V}$ is a nonprincipal ultrafilter and $\mathcal{U} \geq_T \mathcal{V}$, then $\mathcal{V}$ is isomorphic to an ultrafilter of $\vec{\mathcal{W}}$-trees, where $\hat{\mathcal{S}} \setminus \mathcal{S}$ is a well-founded tree, $\vec{\mathcal{W}} = (\mathcal{W}_s : s \in \hat{\mathcal{S}} \setminus \mathcal{S})$, and each $\mathcal{W}_s$ is isomorphic to $\mathcal{D}_0$ or $\mathcal{D}_{i,j}$ for some $(i,j) \in \omega \times \omega$ with $(i,j) \neq (0,0)$.

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Theorem (T., [11])

Suppose that $\mathcal{U}$ is a Ramsey for $\mathcal{H}^2$ ultrafilter on $[T_1 \otimes T_1]$ generated $\mathcal{C} \subseteq \mathcal{H}^2$. If $\mathcal{V}$ is an ultrafilter on ω and $\mathcal{V}$ is Tukey reducible to $\mathcal{U}$ then one of the following holds,

1. $\mathcal{V} \equiv_T \mathcal{D}_{\langle\rangle}$,
2. $\mathcal{V} \equiv_T \mathcal{D}_0$,
3. $\mathcal{V} \equiv_T \mathcal{D}_{0,1}$,
4. $\mathcal{V} \equiv_T \mathcal{D}_{1,0}$ *or*
5. $\mathcal{V} \equiv_T \mathcal{D}_{1,1}$.

Theorem (T., [11])

It is consistent with ZFC that the four-element Boolean algebra appears as an initial Tukey structure.

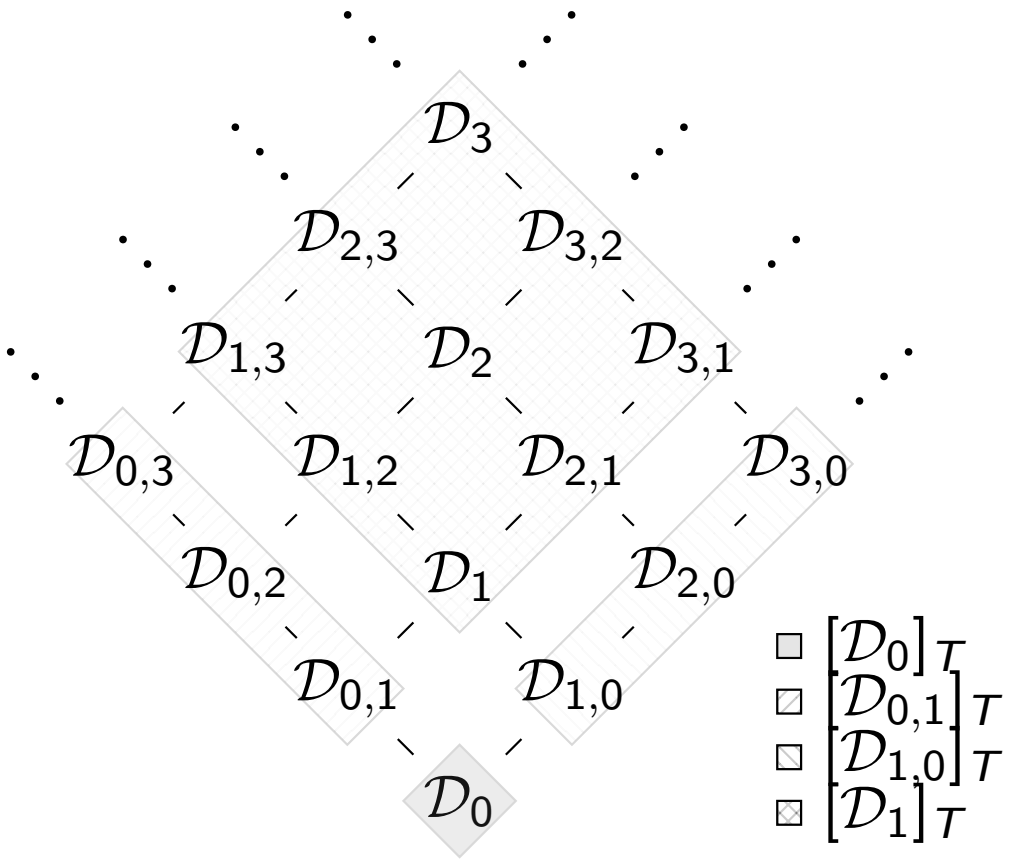

Figure: Rudin-Keisler structure of the p-point ultrafilters within the Tukey types of nonprincipal ultrafilters Tukey reducible to $\mathcal{D}_1$

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- (Dobrinen) What are the possible initial Tukey structures for ultrafilters on a countable base set?

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